

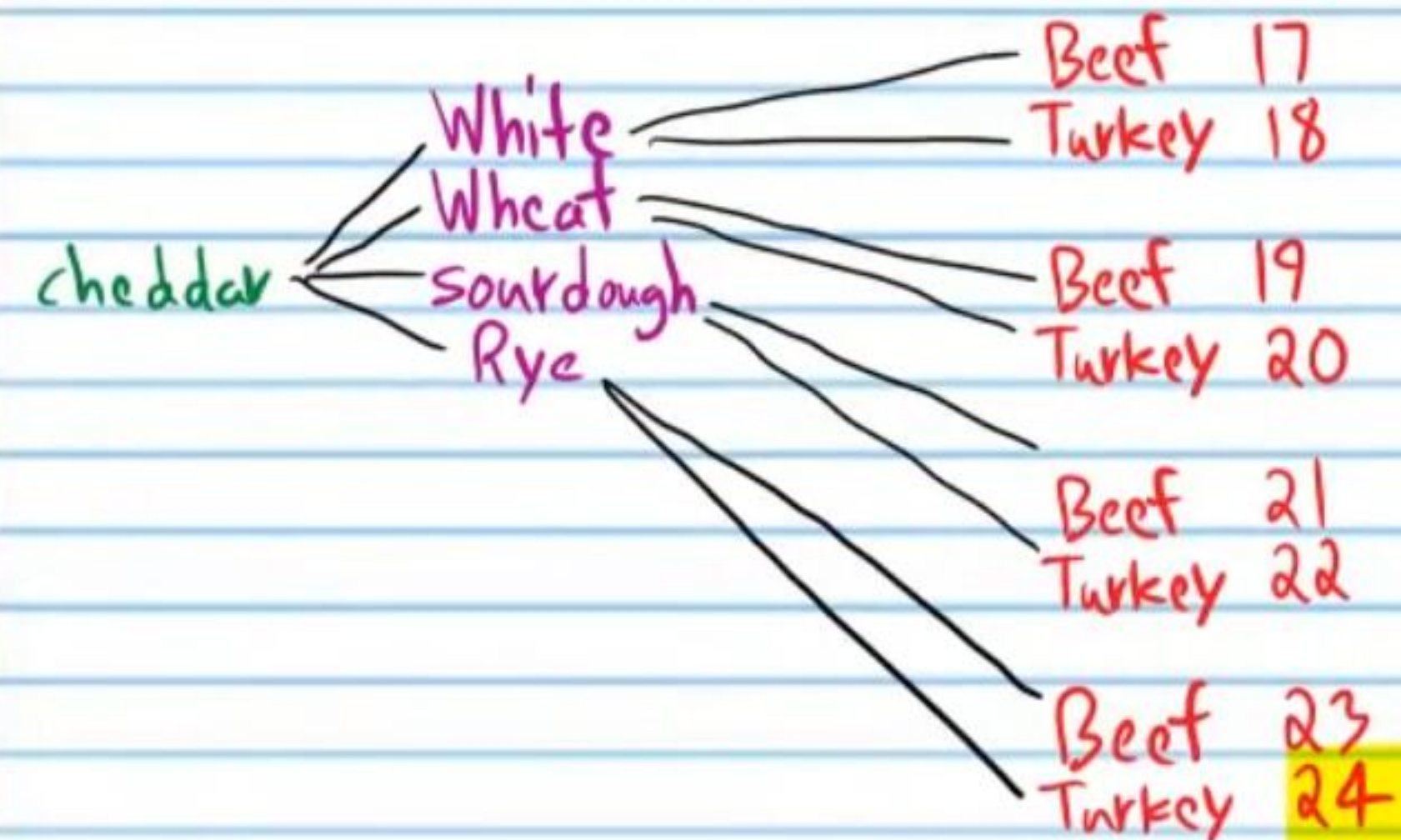
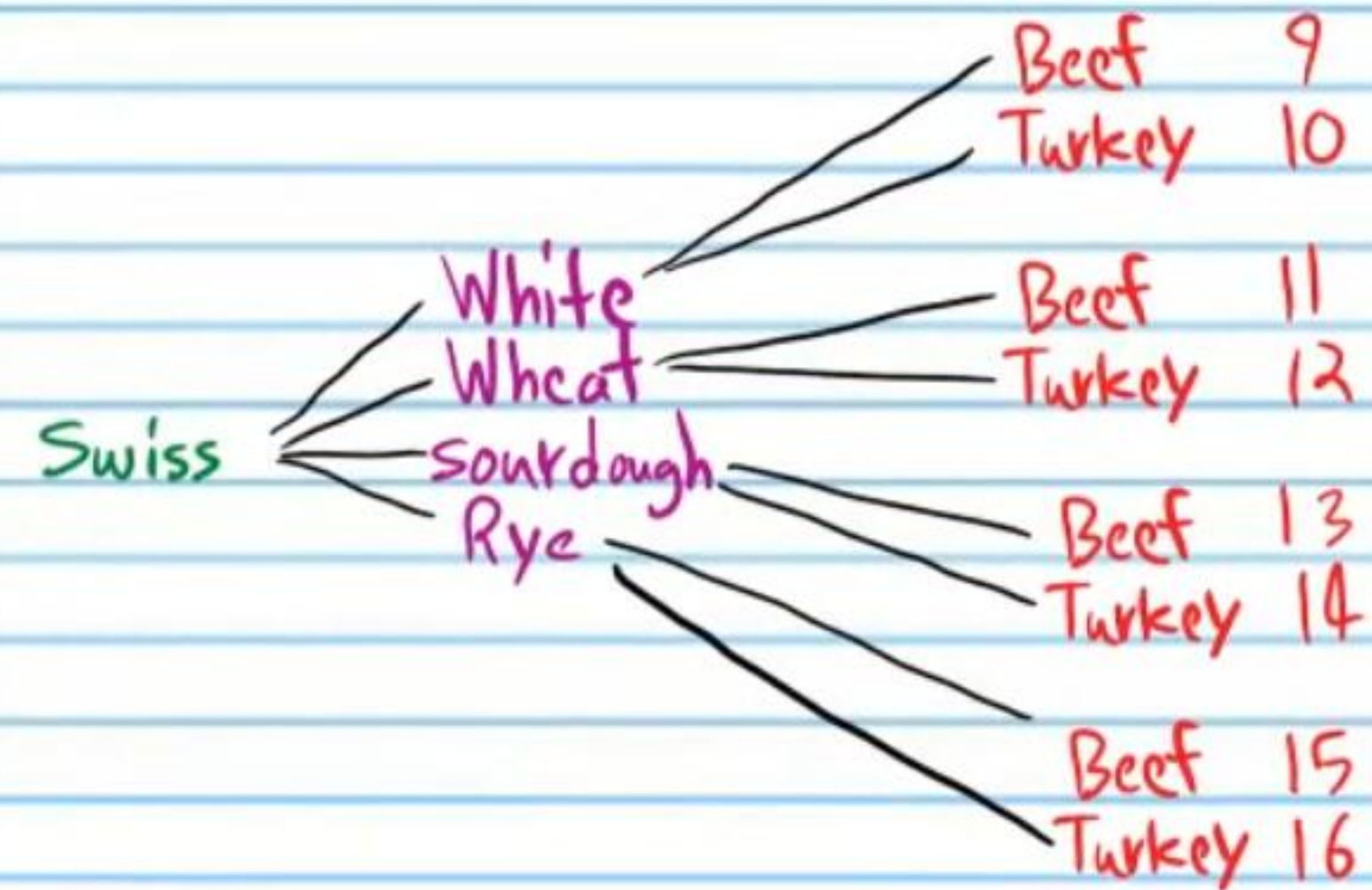
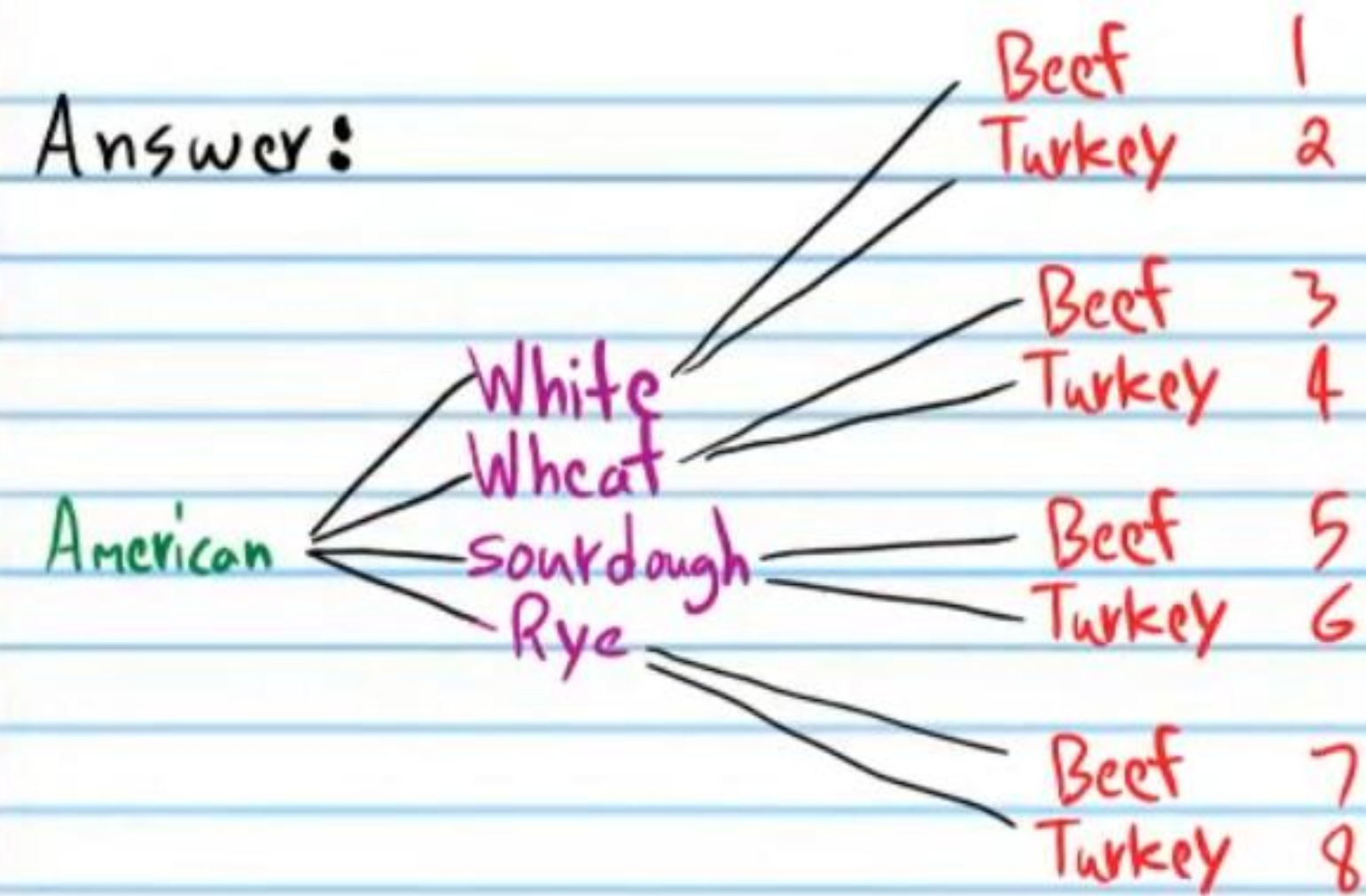
# Counting

## The Multiplication Principle Used for Counting

Example: How many types of sandwiches can you make if you are allowed to choose one type of cheese, one type of bread, and one type of meat from the following menu?

| Cheeses  | Bread     | Meat   |
|----------|-----------|--------|
| American | White     | Beef   |
| Swiss    | Wheat     | Turkey |
| Cheddar  | Sourdough |        |
|          | Rye       |        |

Answer:



$$3 \times 4 \times 2 = 24$$



## Practice

- ① You have 6 different pairs of shoes, 8 different t-shirts, 5 different pairs of pants, and 2 different hats. How many outfit variations are possible?

$$6 \times 8 \times 5 \times 2 = \boxed{480 \text{ outfit variations}}$$

- ② Jill's father took her to the fair. He said that she could take one of her 5 closest friends, she could have cotton candy, either pink or blue, and she would have time to choose only one ride, the Ferris wheel, the merry-go-round, the roller coaster, or the bumper cars. How many outcome variations does Jill have for her day at the fair?

$$5 \times 2 \times 4 = \boxed{40 \text{ outcome variations}}$$

- ③ Ken is buying a new car. He has five choices to make (below). How many outcome variations are possible?

| Engine    | Color | Upholstery | Wheels    | Transmission |
|-----------|-------|------------|-----------|--------------|
| 220cc V-6 | Blue  | Leather    | Spoked    | 4-speed      |
| 243cc V-6 | Red   | Vinyl      | Magnesium | 5-speed      |
| 301cc V-8 | White |            |           | 6-speed      |
| 368cc V-8 | Black |            |           |              |
|           | Green |            |           |              |

$$4 \times 5 \times 2 \times 2 \times 3 = \boxed{240 \text{ outcome variations}}$$

- ④ Alan is buying a new phone. He can choose an Apple, Samsung, or LG. He can choose the Verizon network or T-Mobile. And lastly, he can buy loss insurance or not. How many outcome variations are there?

$$3 \times 2 \times 2 = \boxed{12 \text{ outcome variations}}$$

- ⑤ Amy has the following choices to make regarding her lunch. How many outcome variations are possible for her meal?

| Drink       | Main Course | Side         | Dining   |
|-------------|-------------|--------------|----------|
| Pepsi       | Hamburger   | Apple        | Outside  |
| Water       | Hotdog      | Potato chips | Barstool |
| Dr. Pepper  | Tacos       | Banana       | Booth    |
| Crush       | Pizza       | Orange       |          |
| Mtn. Dew    |             | Cantaloupe   |          |
| Fruit Juice |             |              |          |

$$6 \times 4 \times 5 \times 3 = \boxed{360 \text{ outcome variations}}$$

- ⑥ The Smith family is planning a vacation. They are considering Paris, New York, or Tokyo. They can arrive by plane or on a cruise ship. They can stay in a hotel or with friends that they know in those locations. While they are there, they can rent a car, a truck, or walk to their destinations. And lastly, they can bring their dog or leave him at home. How many outcome variations are possible for their vacation?

$$3 \times 2 \times 2 \times 3 \times 2 = \boxed{72 \text{ outcome variations}}$$



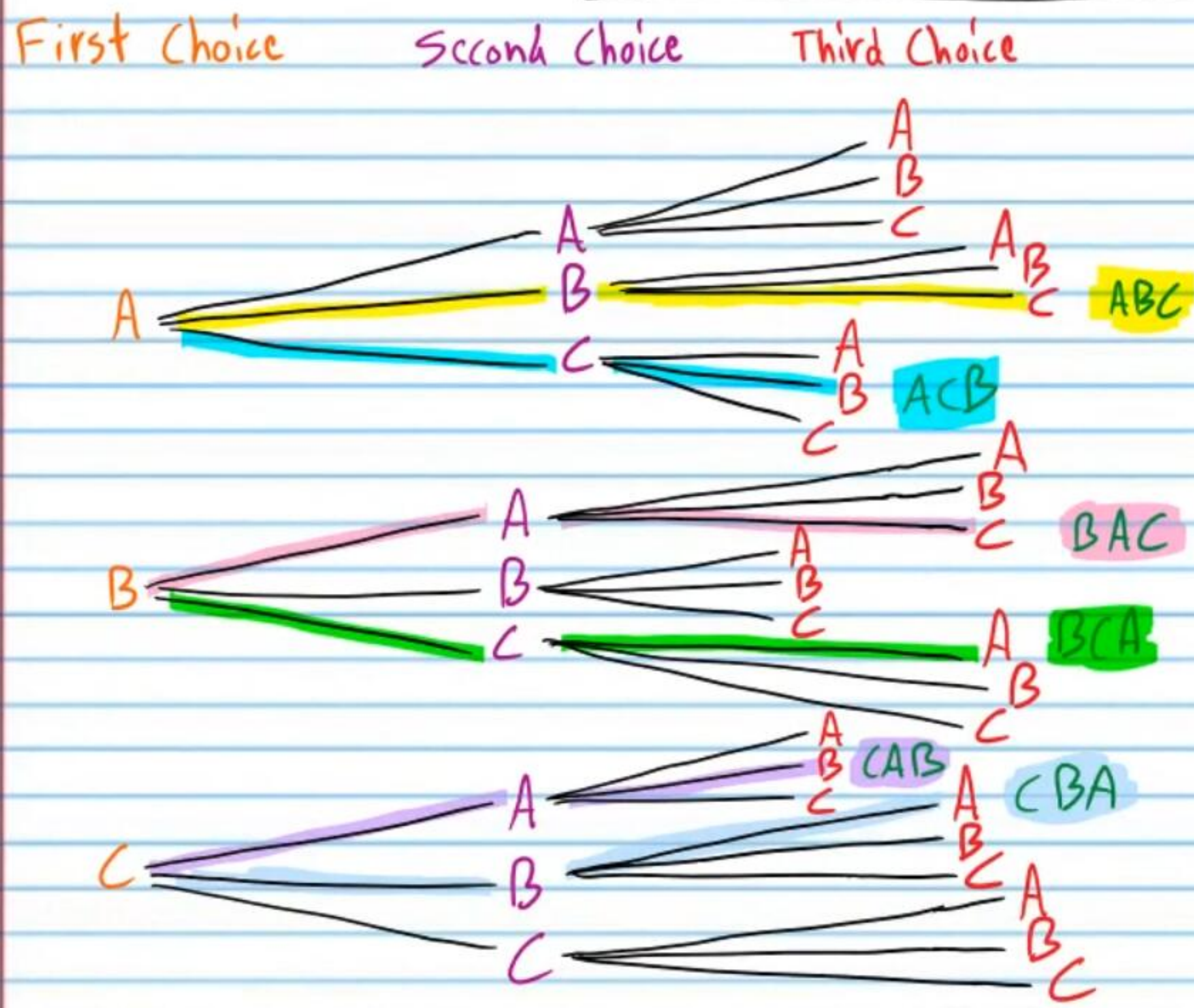
When there are "r" choices and "n" options for each choice, the number of outcome variations is  $n^r$ . All possible permutations (i.e. orders) are counted in this scenario.

Example: Choose a three letter code when repetition of the letters is allowed.

Answer: Three choices,  $r=3$ .

26 letters in the alphabet, 26 options,  $n=26$ .

$$26 \cdot 26 \cdot 26 = 26^3 = \boxed{17,576 \text{ outcome variations}}$$



## Practice

- ⑦ A bank customer must choose a 4-digit pin number. Each digit can be 0-9. How many different outcome variations are there for the customer's pin number? **Zero can be used at any or every place value.**

$$10 \times 10 \times 10 \times 10 = 10^4 = \boxed{10,000 \text{ pin \#s}}$$

- ⑧ RNA is a strand of genetic information made of subunits called nucleotides. Nucleotides are named by the base that they contain: adenine (A), guanine (G), cytosine (C), and uracil (U). How many 15 nucleotide RNA strands are possible?

$$4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 = 4^{15}$$

$$\rightarrow \boxed{1,073,741,824 \text{ strands}}$$

- ⑨ Nobody knows the dates of Kyle, Lauren, or Eddie's birthdays. How many possible outcome variations are there for the dates?

$$365 \times 365 \times 365 = \boxed{48,627,125 \text{ outcome variations}}$$

- ⑩ There are 28 letters in the arabic alphabet. How many ways can 5 letters be permuted (i.e. ordered) if repetition is allowed (i.e. letters can be chosen more than once)?

$$28 \times 28 \times 28 \times 28 \times 28 = 28^5 = \boxed{17,210,368 \text{ ways}}$$



⑪ Computers store information in the form of ones and zeros. How many programs are possible for a group of 29 digits (either ones or zeros)?

$$2 \times 2 \times 2 \times 2 \times 2 \times \dots = 2^{29} = \boxed{536,870,912 \text{ programs}}$$

⑫ There are 20 different types of amino acids used by biological organisms in nature. Amino acids are connected in strands called polypeptides. How many polypeptide variations are possible for a strand that has 12 amino acids?

$$20 \times 20 \times 20 \times 20 \times 20 \times 20 \times 20 \times 20 \times \dots = 20^{12}$$

$$\downarrow$$

$$\boxed{4,096,000,000,000,000 \text{ strands}}$$

Factorial Notation:  $n! = n(n-1)(n-2)(n-3)\dots(1)$

Example:  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Find the following values.

⑬  $4! = 4 \times 3 \times 2 \times 1 = \boxed{24}$

⑭  $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = \boxed{5,040}$

⑮  $8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = \boxed{40,320}$

• ⑯  $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = \boxed{720}$

⑰  $3! = 3 \times 2 \times 1 = \boxed{6}$

⑱  $10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = \boxed{3,628,800}$

When there are "r" choices and "n" options for each choice, and an option cannot be used more than once, the number of outcome variations is  $\frac{n!}{(n-r)!}$ . This expression is abbreviated as  $P(n,r)$  or  $nPr$ .

Example: The Danish alphabet has 29 letters. How many different 6 letter permutations are possible if no letter can be used more than once?

$$P(29,6) = {}_{29}P_6 = 29 \times 28 \times 27 \times 26 \times 25 \times 24$$

$$= \boxed{342,014,400 \text{ permutations}}$$

$$\frac{29!}{(29-6)!} = \frac{29 \times 28 \times 27 \times 26 \times 25 \times 24 \times 23 \times 22 \times 21 \times \dots \times 1}{23 \times 22 \times 21 \times 20 \times \dots \times 1}$$

Practice

- ⑲ There are 27 letters in the Spanish alphabet. How many outcome variations are there for a four letter code if no letter can be used more than once?

$$P(27,4) = {}_{27}P_4 = 27 \times 26 \times 25 \times 24 = \boxed{421,200 \text{ codes}}$$



②② There are 20 different types of amino acids used by biological organisms in nature. Amino acids are connected in strands called polypeptides. How many polypeptide variations are possible for a strand that has 9 amino acids if no amino acid is used more than once?

$$P(20, 9) = {}_{20}P_9 = 20 \times 19 \times 18 \times 17 \times 16 \times 15 \times 14 \times 13 \times 12 \\ = \boxed{60,949,324,800 \text{ strands}}$$

②① Leonard will give one card to each of his five children. He is using a deck with 52 distinct cards. How many outcome variations are possible?

Note: A scenario where Cecelia gets the ace of diamonds is different from a scenario where she gets the 8 of clubs.

$$P(52, 5) = {}_{52}P_5 = 52 \times 51 \times 50 \times 49 \times 48 \\ = \boxed{311,875,200 \text{ outcome variations}}$$

• ②② A bank customer must choose a 4-digit pin number. Each digit can be 0-9, but no number can be used more than once. How many different outcome variations are there for the customer's pin number?

$$P(10, 4) = {}_{10}P_4 = 10 \times 9 \times 8 \times 7 = \boxed{5,040 \text{ outcome variations}}$$

②③ How many 5 letter permutations are possible if a letter cannot be used more than once? There are 26 letters in the English alphabet.

$$P(26, 5) = {}_{26}P_5 = 26 \times 25 \times 24 \times 23 \times 22 \\ = \boxed{7,893,600 \text{ permutations}}$$

②④ Nobody knows the dates of Nicky, Jeff, and Marla's birthdays. How many possible outcome variations are possible for the dates if we know that none of their birthdays are on the same day?

$$P(365, 3) = {}_{365}P_3 = 365 \times 364 \times 363 \\ = \boxed{48,228,180 \text{ outcome variations}}$$

• ②⑤ There are 10 women auditioning for a play that has 6 distinct female characters. How many outcome variations are possible for the female cast if an actor is not allowed to play more than one character?

$$P(10, 6) = {}_{10}P_6 = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \\ = \boxed{151,200 \text{ casts}}$$



②⑥ Four musicians must be chosen to play in a rock band. There will be a lead guitarist, a bass player, a singer, and a drummer. If there are 17 musicians auditioning, and each one is capable of playing any instrument, how many different outcome variations are possible? *Note: A band where Andy is the lead guitarist and Bill plays bass is different from a band where Bill is the lead guitarist and Andy plays bass. Assume that only one musician is allowed to sing.*

$$P(17,4) = {}_{17}P_4 = 17 \times 16 \times 15 \times 14 = \boxed{57,120 \text{ bands}}$$

• ②⑦ 15 students try out for the school baseball team. There are nine distinct positions. How many possible outcome variations are there if only nine students can be chosen? *Note: A team where John pitches is different from a team where Bill pitches.*

$$P(15,9) = {}_{15}P_9 = 15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \\ = \boxed{1,816,214,400 \text{ teams}}$$

②⑧ Investors must choose a 3-person management team. The positions are president, treasurer, and vice president. If the investors can choose from 11 employees of the company, how many outcome variations are possible? *Note: A team where Douglas is the president and Marvin is the vice president is different from a team where Marvin is the president and Douglas is the vice president.*

$$P(11,3) = {}_{11}P_3 = 11 \times 10 \times 9 = \boxed{990 \text{ teams}}$$

• ②⑨ How many different ways can 6 people be lined up?

$$P(6,6) = {}_6P_6 = \underline{6} \times \underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 6! = \boxed{720 \text{ ways}}$$

③⑩ Amanda will put 5 pies on the windowsill to allow them to cool. One is apple, another is lemon, another is blueberry, another is cherry, and the last is pumpkin. How many ways can she line them up?

$$P(5,5) = {}_5P_5 = \underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 5! = \boxed{120 \text{ ways}}$$

• ③⑪ How many different ways can 9 people be lined up?

$$P(9,9) = {}_9P_9 = 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 9! \\ = \boxed{362,880 \text{ ways}}$$

③⑫ Jerry has four different hats that he will wear in the next four days, Monday, Tuesday, Wednesday, and Thursday. If he will wear only one hat per day, and he does not want to wear the same hat twice, how many outcome variations are possible? *Note: If he wears his Angels hat on Monday, this is a different scenario from one where he wears his Angels hat on Tuesday.*

$$P(4,4) = {}_4P_4 = \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 4! = \boxed{24 \text{ outcome variations}}$$



When there are "r" choices and "n" options for each choice, and an option cannot be used more than once, and the order of the things resulting from the choices does not matter, and is not counted, the number of outcome variations is  $\frac{n!}{(n-r)!r!}$ . This expression is abbreviated as  $C(n,r)$  or  $nCr$ .

The things that we are counting are often called "groups" or simply "combinations."

Example: The Danish alphabet has 29 letters. How many different 6 letter combinations are possible if no letter can be used more than once? One combination contains every permutation of the corresponding letters.

$$C(29,6) = {}_{29}C_6 = \frac{29 \times 28 \times 27 \times 26 \times 25 \times 24}{6!} = \boxed{475,020}$$

### Practice

- (33) 12 scientists must be chosen to study the coral reef in Australia. If there are 17 scientists willing to go, how many possible outcome variations are there?

$$C(17,12) = {}_{17}C_{12} = \frac{17 \times 16 \times 15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6}{5!} = \boxed{6,188 \text{ teams}}$$

- (34) There are 10 cats at the pound that need to be rescued. Abbey has a farm and she will pick three of the cats to catch mice. How many outcome variations are possible?

$$C(10,3) = {}_{10}C_3 = \frac{10 \times 9 \times 8}{3!}$$

$$= \boxed{120 \text{ outcome variations}}$$

- (35) There are 32 students in a classroom. If the students are instructed to form groups of 6 people, how many ways can one group be formed?

$$C(32,6) = {}_{32}C_6 = \frac{32 \times 31 \times 30 \times 29 \times 28 \times 27}{6!}$$

$$= \boxed{906,192 \text{ ways}}$$

- (36) There are 27 letters in the Spanish alphabet. How many 4 letter combinations are possible if no letter can be used more than once? One combination contains every permutation of the corresponding letters.

$$C(27,4) = {}_{27}C_4 = \frac{27 \times 26 \times 25 \times 24}{4!}$$

$$= \boxed{17,550 \text{ combinations}}$$



(37) There are 52 cards in a deck. A poker hand is 5 cards. How many poker hands are possible?

$$C(52, 5) = {}_52C_5 = \frac{52 \times 51 \times 50 \times 49 \times 48}{5!}$$
$$= \boxed{2,598,960 \text{ hands}}$$



(38) In the pantry, you find a candy bar, a banana, an apple, an orange, a box of crackers, and a fruit roll-up. You decide to have three of these items for lunch. How many possible outcome variations are there for lunch?

$$C(6, 3) = {}_6C_3 = \frac{6 \times 5 \times 4}{3!}$$

$$= \boxed{20 \text{ outcome variations}}$$